Pricing Impacts of State of Charge Management Options for Electric Storage Resources

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***Abstract*—FERC Order 841 directed the U.S. Independent System Operators (ISOs) to incorporate appropriate changes to their market design and market clearing software to allow for an enhanced participation of electric storage resources (ESRs). A few key aspects included providing ESRs with the option to self-manage their state of charge (SOC) and allowing ESRs to be able to set the wholesale price as both a buyer and seller when it is the marginal resource. This paper contributes to the market design realm by providing detailed theoretical analysis of the impending changes including the potential implications of the different SOC management options on price setting.**

***Index Terms*--Electric storage resource, market design, locational marginal price, price formation, state of charge management.**

NOMENCLATURE

A. Sets

K	Generating facilities (index k); $K^{ESR} \subseteq K$ is the subset of participating electric storage resources.
T	Time periods (index t); $T^{SOC} \subseteq T$ is the subset in which target or desired SOC levels are provided.

B. Parameters (index k denotes facility and t time period)

$c^G_{k,t}$	ESR offer in generator (discharge) mode, $/MWh.
$c^L_{k,t}$	ESR bid in load (charge) mode, $/MWh.
$\overline{D}_t$	System-wide real power demand.
$MaxC_{k,t}$	Maximum charge limit (maximum load), MW.
$MaxD_{k,t}$	Maximum discharge (upper operating) limit, MW.
$SOC^{max}_{k,t}$	Maximum SOC limit, MWh.
$SOC^{min}_{k,t}$	Minimum SOC limit, MWh.
$SSOC_k$	Start (beginning) SOC level, MWh.
$TSOC_k$	Target (end-of-horizon) SOC level, MWh.
η_k	Round-trip efficiency, percentage.

C. Variables (index k denotes facility and t time period)

$G_{k,t}$	Scheduled generation (discharge), MW.
$L_{k,t}$	Scheduled load (charge), MW.
LMP_t	Dual variable on real power balance constraint.
$SOC_{k,t}$	State of charge (energy) level, MWh.
$\epsilon_{k,t}$	Dual variable on upper bound (maximum discharge) capacity constraint.
$\theta_{k,t}$	Dual variable on lower bound (maximum charge) capacity constraint.
ξ_k	Dual variable on beginning SOC constraint.
$\gamma_{k,t}$	Dual variable on SOC throughput constraints.
$\psi_{k,t}$	Dual variable on desired SOC constraints.

I. INTRODUCTION

Many regions throughout the world anticipate large-scale variable energy resource (VER) penetration increases on their system. Electric storage technology is an emerging technology that has been widely viewed as a further enabler of high penetrations of VERs, such as wind and solar photovoltaic. Energy storage has seen substantial growth in recent years. While many energy storage systems have been used for distribution system asset deferral and for customer-focused application (e.g., home backup), a number of technologies have also been participating in wholesale electricity markets for bulk power system services, including energy, ancillary services (AS), and capacity. For instance, pumped storage hydro (PSH) resources have been around for decades providing bulk power system services [1].

Battery technologies, particularly lithium ion technology, has seen a substantial decrease in the cost [2]. Some batteries and flywheels have been participating in certain AS markets in some Independent System Operators (ISOs) for several years now based on anticipated attractive revenue possibilities in those markets [3]. Particularly, battery technologies have primarily participated in regulating reserve AS markets, providing fast response in both upward and downward directions to support frequency and area control error within the ISO footprint. This type of participation in regulation markets has allowed ESRs to manage their state of charge (SOC) due to the continuous bi-directional use of charging and discharging energy. Regulation is typically the highest priced AS making it a desirable service for these technologies to earn the most revenue. Historically, these technologies have generally not provided energy or other AS. This can be due to both eligibility requirements and barriers (limited ability to participate via the existent participation structure) as well as economic reasons (they may not make as much revenue as made in regulating reserve markets).

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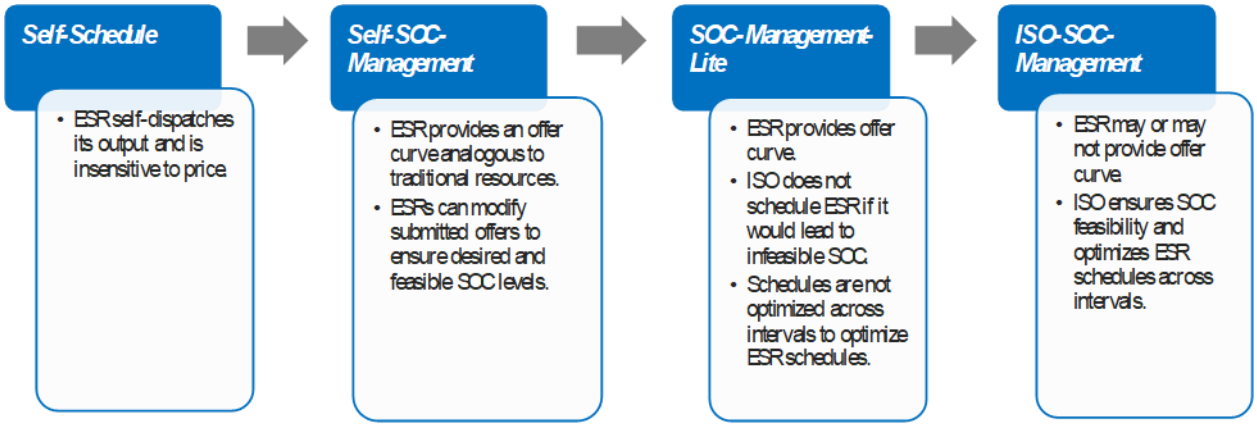


Figure I-1. State of Charge Management Options for ESRs

Numerous states in the United States have initiated or are planning on initiating targets for energy storage deployment [4], [5]. Furthermore, the Federal Energy Regulatory Commission (FERC) directed the U.S. ISOs/RTOs to make changes to their market design and market clearing software to allow for the enhanced participation of electric storage resources (ESRs) through its Order 841 [6]. In addition, all regions inside and outside of FERC's jurisdiction (e.g., ERCOT, IESO, and AESO) have interested stakeholders requesting new ways for ESRs to participate within their existing and evolving market design. Many of these requests may require substantial changes to the market clearing software. The potential changes that are being proposed in different regions are discussed in great detail in [7]. One of the main aspects of the Order was to provide ESRs with the option to self-manage their SOC and not impose ISO management of SOC as a requirement. Order 841 also necessitates that ESRs must be able to set the wholesale price as both a buyer and seller when it is a marginal resource in a manner that is similar to how other resources set prices. In other words, when providing offers as a generator or demand response simultaneously and getting selected as one or the other, the ESR can set price in the energy market just as generators and dispatchable demand can when it offers its power range flexibly. The aforesaid components of the FERC Order are of high relevance to the research that is conducted in this paper. References [8] and [9] provide both formulaic and quantitative insights into the ways in which the different SOC management (SOCM) options can be implemented. Results provide key insights on a few potential outcomes including economic efficiency, reliability and SOC feasibility outcomes. This paper further evaluates the potential implications of the different SOCM options on price setting including price formation with ESRs as marginal resources, i.e., how/when ESRs can set price. The primary goal is to assess alternative wholesale electricity market designs that are currently being proposed for incorporating ESRs, through comprehensive modeling and analysis, in addition to understanding the impacts on electricity market outcomes. This will help provide key insights for all industry participants.

The paper is organized as follows. Section II investigates the different SOCM design options, including the market modeling options and the mathematical formulations. A theoretical analysis of the potential implications of the different SOCM options on price setting is provided in Section III.

II. Evaluation of State of Charge Management Options

There are numerous ways in which the characteristics of ESRs can be incorporated into electricity market clearing software. It is important to understand the implications of the different design options that are currently being proposed and those that are close to implementation. The main characteristic of ESRs that is explored in this research is its SOC limitation – the fact that an ESR can only supply continuous energy for a limited amount of time, after which it is fully depleted and must charge again to replenish its SOC. This characteristic has a significant impact on the ability and quantity of energy, capacity, and ancillary services that the ESR is able to sell. Figure I.1. illustrates the different SOCM design options. The options vary primarily based on the responsibility of the entity that schedules the resource and manages its SOC. The options moving right require more management and scheduling decisions from the ISO while the options moving left require more management and scheduling decisions from the ESR asset owner.

The requirement that the ISOs must allow ESRs to self-manage its SOC as part of ESR participation within the markets means that it would be the ESR owner's responsibility to ensure that its SOC levels are feasible based on how it offers into the markets. The ISO can have an option to manage the SOC, but it must be up to the ESR owners to use that option. As part of the self-management requirement, ESRs would be subject to financial penalties for uninstructed deviation (when actual energy is different from ISO-scheduled energy) when managing their SOC and not providing energy schedules due to running out of SOC. Note that while Self-SOCM was required to be allowed as an option, FERC also required that a number of physical and operational characteristics must be accounted for in the ESR participation model, through bidding parameters or otherwise (including telemetry). This includes: SOC, maximum and minimum SOC limits, maximum and minimum charge and discharge limits, maximum and minimum charge and run times, and charge and discharge ramp rates. This Section describes the market modeling and the mathematical formulation of the different ways to allow ESR participation in wholesale electricity markets.

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A. Design Modifications for Self-SOC-Management

In this option, ESR owners provide cost/quantity offer curves that, to the best ability of the owner, lead to desired and feasible SOC level at all times without the need for explicit ISO intervention. The offer curve can be a piecewise linear monotonically increasing price-quantity pair with the values reflecting the price at which the ESR is willing to pay when in the negative part of its curve, and the price at which it is willing to sell when in the positive part of its curve. SOC and related parameters are not needed by the ISO but may still be monitored for situational awareness or emergency reasons, e.g., PJM's proposal. Thus, in Self-SOCM, the ISO does not explicitly include constraints that relate to SOC, e.g., SOC throughput calculation, minimum and maximum SOC restrictions, target/desired SOC level to be attained at the end of the operating horizon, in its market auction model. A simplistic representation of an ISO's day-ahead security-constrained economic dispatch (DASCED) problem, wherein the system's ESRs are required to self-manage their SOC is detailed below. For the sake of simplicity and ease of understanding, the DASCED problem formulation is simplified by ignoring AS and assuming a single segment *each* to represent the charging and discharging decisions instead of multiple piecewise linear segments. Future phases of this research will evaluate price formation while considering AS.

$$\underset{G_{k,t},L_{k,t}}{\text{Minimize:}} \sum_{k,t}\left(c^{G}_{k,t}G_{k,t} - c^{L}_{k,t}L_{k,t}\right) \quad (1)$$

Subject to:

$$-\left(G_{k,t} - L_{k,t}\right) \geq -MaxD_{k,t} \qquad \left(\epsilon_{k,t}\right) \qquad \forall k \in K^{ESR}, t \quad (2)$$

$$\left(G_{k,t} - L_{k,t}\right) \geq -MaxC_{k,t} \qquad \left(\theta_{k,t}\right) \qquad \forall k \in K^{ESR}, t \quad (3)$$

$$\sum_{k}\left(G_{k,t} - L_{k,t}\right) = \overline{D}_{t} \qquad \left(LMP_{t}\right) \qquad \forall t \quad (4)$$

$$G_{k,t} \geq 0, L_{k,t} \geq 0.$$

The objective, (1), is to maximize social welfare given the offers and bids from generating and load entities, respectively. Constraints (2) and (3) impose maximum discharge and charge limitations on the real power scheduled from ESRs, respectively. Constraint (4) ensures system-wide power balance between generation and demand. Note that the aforesaid formulation only includes specific equations that are related to the subset of ESRs, K^{ESR}, for the sake of discussion and illustration purposes; however, in actual practice, as ISO's DASCED formulation will include additional equations that relate to other resource types and system constraints.

B. Design Modifications for SOC-Management-Lite

The SOCM-Lite option is being proposed for implementation by majority of the ISOs (excepting PJM) and may be a response to the two statements in Order 841 mentioning that ISO-SOCM is not required but that the ISO must consider an ESR's SOC and SOC limits as parameters that account for the ESR's operational characteristics. The ESR uses a similar offer curve as in the Self-SOCM option, but in addition the ISO monitors the SOC and ensures that the SOC levels are feasible. The ISO does not ensure that the SOC is optimal or set to desired levels. The economic selection of an ESR is determined primarily based on the bidding strategy of the ESR owner; however, infeasible solutions are not allowed.

$$\underset{G_{k,t},L_{k,t},SOC_{k,t}}{\text{Minimize:}} \sum_{k,t}\left(c^{G}_{k,t}G_{k,t} - c^{L}_{k,t}L_{k,t}\right) \quad (1)$$

Subject to:

Include constraints (2) through (4).

$$SOC_{k,0} = SSOC_{k} \qquad \left(\xi_{k}\right) \qquad \forall k \in K^{ESR} \quad (5)$$

$$-SOC_{k,t} \geq -SOC^{max}_{k,t} \qquad \left(\overline{\Upsilon_{k,t}}\right) \qquad \forall k \in K^{ESR}, t \quad (6)$$

$$SOC_{k,t} \geq SOC^{min}_{k,t} \qquad \left(\underline{\Upsilon_{k,t}}\right) \qquad \forall k \in K^{ESR}, t \quad (7)$$

$$SOC_{k,t} = SOC_{k,t-1} - \frac{1}{\eta^{G}_{k}}G_{k,t} + \eta^{L}_{k}L_{k,t} \qquad \left(\Upsilon_{k,t}\right) \qquad \forall k \in K^{ESR}, t \quad (8)$$

$$G_{k,t} \geq 0, L_{k,t} \geq 0, SOC_{k,t} \geq 0.$$

In other words, in the SOCM-Lite option, the ISO ensures SOC feasibility in its market auction models by explicitly including constraints that bound the SOC level of an ESR to be below its maximum limit, (6), and above its minimum limit, (7). The SOC level is calculated for every interval, (8), based on the amount that is charged and discharged, and a round-trip efficiency value, which reduces the amount of energy that can be stored from what is charged from the grid. The offer curve can be selected based on the ESR's bidding strategy to maximize profit; however, the ESR's bidding strategy itself no longer needs to attempt to also ensure feasibility with regard to SOC limitations, which was otherwise necessary when developing an ESR's offer strategy in the Self-SOCM option.

C. Design Modifications for ISO-SOC-Management

In this option, the ISO monitors current SOC, anticipated SOC and other related ESR parameters (e.g., round-trip efficiency), and provides schedules that explicitly lead to a desired and feasible SOC level at all times. Offer curves are not necessarily needed, though can potentially also be combined (e.g., to incorporate degradation costs). This option conducts a time-coupled multi-period optimization, which optimizes the ESR in a manner that minimizes the system's production costs. Since the ESR's effective costs are a result of the optimization rather than based on exogenous offer patterns from the ESRs, this option should in theory provide the most economically efficient option, assuming that the anticipated conditions used by the ISO are accurate. However, it is important to note that in non-convex problems, cost minimization may not always lead to profit maximization. Given that the DASCUC problem is highly non-convex, it is possible that while the theoretic least cost solution may result from this option, it is not guaranteed that it will lead to the best solution for ESR owners in terms of maximal profit. However, since times of greatest need align with higher prices, it is still possible.

For the purpose of subsequent discussions, it is assumed that the ESR owners do not submit offers in the ISO-SOCM option. This option requires additional information to allow the ESR to get to a desired SOC level at the end of the optimization horizon, and to avoid myopic decisions that may empty out the ESR without leaving any stored energy for the subsequent day. One option is to have a single desired SOC level that is used for the last interval of the optimization horizon, (9). Without this option, the SOC would likely be empty at the end of any optimization horizon as the objective function would use all available stored energy to reduce costs. An additional option would be to include desired SOC levels throughout an optimization horizon, i.e., at different time instances within the optimization horizon to provide the ESR with more flexibility to be used in a way it may want to be used. A third option would

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be to allow the ESR to offer a "End of Horizon SOC Value" (c^{SOC}) for the end of the optimization horizon. Each of the three options has different advantages but allows for better use of the ESR with ISO-SOCM by letting the ESR provide information on how much energy the ISO should leave at the end of the horizon, either a specific level, or a level that will depend on the overall costs for the time horizon.

1) ***Option 1****: End of Horizon Desired SOC Level*

$$\underset{G_{k,t},L_{k,t},SOC_{k,t}}{\text{Minimize:}} \sum_{k\notin K^{ESR},t}\left(c^{G}_{k,t}G_{k,t}-c^{L}_{k,t}L_{k,t}\right) \quad (1a)$$

Subject to:

Include constraints (2) through (8).

$$SOC_{k,end}=TSOC_{k} \qquad (\psi_k) \qquad \forall k\in K^{ESR} \quad (9)$$

2) ***Option 2****: Multiple Desired SOC Levels Throughout the Horizon*

Same objective as (1a). Include constraints (2) through (8).

$$SOC_{k,t}=TSOC_{k,t} \qquad \left(\psi_{k,t}\right)\ \forall k\in K^{ESR}, t\in T^{SOC} \quad (10)$$

3) ***Option 3****: End of Horizon SOC Value Offer*

$$\underset{G_{k,t},L_{k,t},SOC_{k,t}}{\text{Minimize:}} \sum_{k\notin K^{ESR},t}\left(c^{G}_{k,t}G_{k,t}-c^{L}_{k,t}L_{k,t}\right)- \sum_{k\in K_{ESR}} c^{SOC}_{k}SOC_{k,end} \quad (1b)$$

Subject to:

Include constraints (2) through (8).

III. Price Formation

FERC Order 841 provides that ESRs must be able to set the wholesale energy price as both a buyer and seller when participating in the wholesale market specifically when they are the marginal energy resource. Thus, they may set the price when receiving energy (charging) when they are the marginal buyer — meaning the next increment of fixed load increases will result in a reduction in the resources charging load— and the price would be set to the bid the ESR provided as its willingness to buy energy or equivalent. Other price setting criteria for existing resources still apply: when ESRs self-schedule their output they cannot set price; and when ESRs are scheduled at a minimum output, they also cannot set price. The energy purchased from the wholesale market must later be sold back to the market at a wholesale price. This also means that the nodal price must be used for both selling and buying power to prevent gaming between nodal and zonal pricing differences. ESRs must also be able to set the price for AS markets and capacity markets when marginal; however, this paper focuses on ESRs setting the price in energy markets when marginal. Note that all ISOs have proposed that ESRs are eligible to: 1) set wholesale prices in all markets when marginal, and 2) purchase/sell at the wholesale nodal locational marginal prices (LMPs). However, PJM has a unique modeling option for PSH resources; PSH plants can participate in PJM's energy markets either by self-scheduling or by using its pumped hydro optimizer (DA only) but are required to submit zero-cost offer curves, are disallowed from setting wholesale prices and offering negative dispatchable range.

The different SOCM options have far reaching implications on everything from price setting, market settlements, market mitigation, and computational efficiency. This Section provides a theoretical explanation of the impact that ESRs have on setting DA market clearing prices (MCPs) for energy with the changes that are incorporated in the different SOCM options. Lagrangian relaxation is used to enable a theoretical analysis of the impending changes. The goal is to investigate the effect of newly proposed changes in greater detail. Note that, typically, MCPs are set to be equal to the marginal cost of the marginal resource for providing energy or a specific AS.

A. Price Setting Results: Self-SOC-Management

The Lagrangian relaxation of the constrained optimization problem described in Section II A is provided below:

$$\mathcal{L}=\sum_{k,t}\left(c^{G}_{k,t}G_{k,t}-c^{L}_{k,t}L_{k,t}\right)-\sum_{k,t}\epsilon_{k,t}\left[-\left(G_{k,t}-L_{k,t}\right)+MaxD_{k,t}\right]-\sum_{k,t}\theta_{k,t}\left[\left(G_{k,t}-L_{k,t}\right)+MaxC_{k,t}\right]-\sum_{t}LMP_{t}\left[\sum_{k}\left(G_{k,t}-L_{k,t}\right)-\overline{D_{t}}\right] \quad (11)$$

$$\frac{\partial\mathcal{L}}{\partial G_{k,t}}=c^{G}_{k,t}+\epsilon_{k,t}-\theta_{k,t}-LMP_{t}=0$$

$$LMP_{t}=c^{G}_{k,t}+\epsilon_{k,t}-\theta_{k,t}, \quad (12)$$

$$\frac{\partial\mathcal{L}}{\partial L_{k,t}}=-c^{L}_{k,t}-\epsilon_{k,t}+\theta_{k,t}+LMP_{t}=0$$

$$LMP_{t}=c^{L}_{k,t}+\epsilon_{k,t}-\theta_{k,t}. \quad (13)$$

The LMP can be calculated based on (12) and (13). For additional piecewise linear segments of offer curve, capacity constraints for each segment (i) would be included with an associated dual variable (e.g., $\epsilon_{k,t,i}$ and $\theta_{k,t,i}$). In Self-SOCM, the ISO can choose to use only one scheduling decision variable to represent both charging and discharging decisions simultaneously. When this is the case, the price setting for when ESRs are marginal are no different from when other traditional resources set the price with the only exception being that the ESR can set the price when charging (as a negative injection). When discharging, the ESR can set the price at its location equal to its offer for the particular segment of its piecewise offer curve if it is marginal. When charging, it can also set the price at its location equal to its bid to buy for the particular segment of its piecewise offer curve if it is marginal. When a resource has minimum discharge or charge limits and is scheduled at either of those limits it may not set the price. If the ESR is ramp constrained, it also may not set the price even when scheduled at a point within its offered segments.

B. Price Setting Results: SOC-Management-Lite

Price setting for ESRs that are marginal in the SOCM-Lite option is similar to that in the Self-SOCM option. The primary difference is that when an ESR is limited by its SOC constraint, yet still providing within an offer segment (i.e., not limited at either bound of the segment capacity constraints), it will not be eligible to set the MCP. The SOC constraint can be considered similar to a *modified capacity constraint* for a generator, such that the maximum block it can provide reduces when SOC limits are binding. This is demonstrated mathematically. The Lagrangian relaxation of the constrained optimization problem described in Section II B is provided below:

$$\mathcal{L}=\sum_{k,t}\left(c^{G}_{k,t}G_{k,t}-c^{L}_{k,t}L_{k,t}\right)-\sum_{k,t}\epsilon_{k,t}\left[-\left(G_{k,t}-L_{k,t}\right)+MaxD_{k,t}\right]-\sum_{k,t}\theta_{k,t}\left[\left(G_{k,t}-L_{k,t}\right)+MaxC_{k,t}\right]-\sum_{t}\underline{LMP_{t}}\left[\sum_{k}\left(G_{k,t}-L_{k,t}\right)-\overline{D_{t}}\right]-\sum_{k}\xi_{k}\left(SOC_{k,0}-SSOC_{k}\right)-\sum_{k,t}\overline{\Upsilon_{k,t}}\left(-SOC_{k,t}+SOC^{max}_{k,t}\right)-\sum_{k,t}\underline{\Upsilon_{k,t}}\left(SOC_{k,t}-\right.$$

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$$SOC_{k,t}^{min}\big) - \sum_{k,t}\gamma_{k,t}\left[SOC_{k,t} - \left(SOC_{k,t-1} - \frac{1}{\eta_k^G}G_{k,t} + \eta_k^L L_{k,t}\right)\right] \quad (14)$$

$$\frac{\partial \mathcal{L}}{\partial G_{k,t}} = c_{k,t}^G + \epsilon_{k,t} - \theta_{k,t} - LMP_t - \frac{1}{\eta_k^G}\Upsilon_{k,t} = 0$$

$$LMP_t = c_{k,t}^G + \epsilon_{k,t} - \theta_{k,t} - \frac{1}{\eta_k^G}\Upsilon_{k,t}, \quad (15)$$

$$\frac{\partial \mathcal{L}}{\partial L_{k,t}} = -c_{k,t}^L - \epsilon_{k,t} + \theta_{k,t} + LMP_t + \eta_k^L\Upsilon_{k,t} = 0$$

$$LMP_t = c_{k,t}^L + \epsilon_{k,t} - \theta_{k,t} - \eta_k^L\Upsilon_{k,t}, \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial SOC_{k,t}} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} - \Upsilon_{k,t} + \Upsilon_{k,t+1} = 0 \quad (17)$$

$$\Upsilon_{k,t} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} + \Upsilon_{k,t+1}, \quad (18)$$

$$\Upsilon_{k,0} = -\xi_k + \overline{\Upsilon_{k,0}} - \underline{\Upsilon_{k,0}} + \Upsilon_{k,1}. \quad (19)$$

The LMP can be calculated based on (15) and (16). If the ESR is not operating at its maximum discharge limit or at its maximum charge limit, the shadow prices for constraints (2) and (3) will be zero; consequently, the LMP in this case will be equal to the summation of the marginal cost (or the corresponding submitted offer) of the ESR and the dual variable (LaGrange multiplier) of the SOC constraint weighted by the discharging or charging efficiency (or round-trip efficiency) correspondingly. If the SOC level of the ESR is within its SOC bounds (i.e., strictly less than its maximum SOC limit and strictly more than its minimum SOC limit), the shadow prices for (6) and (7) will be zero. The dual variable of the SOC constraint ($\Upsilon_{k,t}$) is impacted by the dual variables of future SOC constraints for future intervals from (18). Note that the shadow price of the SOC constraint in the last interval ($\Upsilon_{k,end}$) will be different from (18) and will only consist of the first two components because there are no further future intervals in the optimization horizon. In addition, when the SOC limit is binding, the ESR will not set the price even when scheduled within its dispatch segment blocks (i.e., not bounded by one of its piecewise segments).

C. Price Setting Results: ISO-SOC-Management

The ISO-SOCM options do not require offers from the ESRs; thus, the objectives (1a) and (1b) exclude the subset of ESRs when maximizing the market surplus. However, if ESRs still elect to submit offers, the corresponding offers can be included in the objective. Thus, when partially differentiating the Lagrangian with respect to an ESR's discharging and charging scheduling decisions, the corresponding offers do not show up in the LMP equations as before. The Lagrangian relaxation of the constrained optimization problems described in II C are provided below.

1) ***Option 1:*** *End of Horizon Desired SOC Level*

$$\mathcal{L} = \sum_{k\notin K_{ESR},t}\left(c_{k,t}^G G_{k,t} - c_{k,t}^L L_{k,t}\right) - \sum_{k,t}\epsilon_{k,t}\left[-\left(G_{k,t} - L_{k,t}\right) + MaxD_{k,t}\right] - \sum_{k,t}\theta_{k,t}\left[\left(G_{k,t} - L_{k,t}\right) + MaxC_{k,t}\right] - \sum_t LMP_t\left[\sum_k\left(G_{k,t} - L_{k,t}\right) - \overline{D_t}\right] - \sum_k \xi_k\left(SOC_{k,0} - SSOC_k\right) - \sum_{k,t}\overline{\Upsilon_{k,t}}\left(-SOC_{k,t} + SOC_{k,t}^{max}\right) - \sum_{k,t}\underline{\Upsilon_{k,t}}\left(SOC_{k,t} - SOC_{k,t}^{min}\right) - \sum_{k,t}\Upsilon_{k,t}\left[SOC_{k,t} - \left(SOC_{k,t-1} - \frac{1}{\eta_k^G}G_{k,t} + \eta_k^L L_{k,t}\right)\right] - \sum_k \psi_k\left(SOC_{k,end} - TSOC_k\right) \quad (20)$$

$$\frac{\partial \mathcal{L}}{\partial G_{k,t}} = \epsilon_{k,t} - \theta_{k,t} - LMP_t - \frac{1}{\eta_k^G}\Upsilon_{k,t} = 0$$

$$LMP_t = \epsilon_{k,t} - \theta_{k,t} - \frac{1}{\eta_k^G}\Upsilon_{k,t}, \quad (21)$$

$$\frac{\partial \mathcal{L}}{\partial L_{k,t}} = -\epsilon_{k,t} + \theta_{k,t} + LMP_t + \eta_k^L\Upsilon_{k,t} = 0$$

$$LMP_t = \epsilon_{k,t} - \theta_{k,t} - \eta_k^L\Upsilon_{k,t}, \quad (22)$$

$$\frac{\partial \mathcal{L}}{\partial SOC_{k,t}} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} - \Upsilon_{k,t} + \Upsilon_{k,t+1} = 0 \quad (23)$$

$$\Upsilon_{k,t} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} + \Upsilon_{k,t+1}, \quad (24)$$

$$\Upsilon_{k,0} = -\xi_k + \overline{\Upsilon_{k,0}} - \underline{\Upsilon_{k,0}} + \Upsilon_{k,1}, \quad (25)$$

$$\gamma_{k,end} = \overline{\Upsilon_{k,end}} - \underline{\Upsilon_{k,end}} - \psi_k. \quad (26)$$

2) ***Option 2****: Multiple Desired SOC Levels Throughout the Horizon*

$$\mathcal{L} = \sum_{k\notin K_{ESR},t}\left(c_{k,t}^G G_{k,t} - c_{k,t}^L L_{k,t}\right) - \sum_{k,t}\epsilon_{k,t}\left[-\left(G_{k,t} - L_{k,t}\right) + MaxD_{k,t}\right] - \sum_{k,t}\theta_{k,t}\left[\left(G_{k,t} - L_{k,t}\right) + MaxC_{k,t}\right] - \sum_t LMP_t\left[\sum_k\left(G_{k,t} - L_{k,t}\right) - \overline{D_t}\right] - \sum_k \xi_k\left(SOC_{k,0} - SSOC_k\right) - \sum_{k,t}\overline{\Upsilon_{k,t}}\left(-SOC_{k,t} + SOC_{k,t}^{max}\right) - \sum_{k,t}\underline{\Upsilon_{k,t}}\left(SOC_{k,t} - SOC_{k,t}^{min}\right) - \sum_{k,t}\Upsilon_{k,t}\left[SOC_{k,t} - \left(SOC_{k,t-1} - \frac{1}{\eta_k^G}G_{k,t} + \eta_k^L L_{k,t}\right)\right] - \sum_{k,t\in T^{soc}}\psi_{k,t}\left(SOC_{k,t} - TSOC_{k,t}\right) \quad (27)$$

Same LMPs as ISO-SOCM Option 1, i.e., (21) and (22).

$$\frac{\partial \mathcal{L}}{\partial SOC_{k,t}} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} - \Upsilon_{k,t} + \Upsilon_{k,t+1} - \psi_{k,t:t\in T^{soc}} = 0 \quad (28)$$

$$\Upsilon_{k,t} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} + \Upsilon_{k,t+1} - \psi_{k,t:t\in T^{soc}}, \quad (29)$$

$$\Upsilon_{k,0} = -\xi_k + \overline{\Upsilon_{k,0}} - \underline{\Upsilon_{k,0}} + \Upsilon_{k,1}, \quad (30)$$

$$\gamma_{k,end} = \overline{\Upsilon_{k,end}} - \underline{\Upsilon_{k,end}} - \psi_{k,end:end\in T^{soc}}. \quad (31)$$

3) ***Option 3****: End of Horizon SOC Value Offer*

$$\mathcal{L} = \sum_{k\notin K_{ESR},t}\left(c_{k,t}^G G_{k,t} - c_{k,t}^L L_{k,t}\right) - \sum_{k\in K_{ESR}} c_k^{SOC} SOC_{k,end} - \sum_{k,t}\epsilon_{k,t}\left[-\left(G_{k,t} - L_{k,t}\right) + MaxD_{k,t}\right] - \sum_{k,t}\theta_{k,t}\left[\left(G_{k,t} - L_{k,t}\right) + MaxC_{k,t}\right] - \sum_t LMP_t\left[\sum_k\left(G_{k,t} - L_{k,t}\right) - \overline{D_t}\right] - \sum_k \xi_k\left(SOC_{k,0} - SSOC_k\right) - \sum_{k,t}\overline{\Upsilon_{k,t}}\left(-SOC_{k,t} + SOC_{k,t}^{max}\right) - \sum_{k,t}\underline{\Upsilon_{k,t}}\left(SOC_{k,t} - SOC_{k,t}^{min}\right) - \sum_{k,t}\Upsilon_{k,t}\left[SOC_{k,t} - \left(SOC_{k,t-1} - \frac{1}{\eta_k^G}G_{k,t} + \eta_k^L L_{k,t}\right)\right] \quad (32)$$

Same LMPs as ISO-SOCM Option 1, i.e., (21) and (22).

$$\frac{\partial \mathcal{L}}{\partial SOC_{k,t}} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} - \Upsilon_{k,t} + \Upsilon_{k,t+1} = 0 \quad (33)$$

$$\Upsilon_{k,t} = \overline{\Upsilon_{k,t}} - \underline{\Upsilon_{k,t}} + \Upsilon_{k,t+1}, \quad (34)$$

$$\Upsilon_{k,0} = -\xi_k + \overline{\Upsilon_{k,0}} - \underline{\Upsilon_{k,0}} + \Upsilon_{k,1}, \quad (35)$$

$$\frac{\partial \mathcal{L}}{\partial SOC_{k,end}} = -c_k^{SOC} + \overline{\Upsilon_{k,end}} - \underline{\Upsilon_{k,end}} - \Upsilon_{k,end} = 0 \quad (36)$$

$$\Upsilon_{k,end} = -c_k^{SOC} + \overline{\Upsilon_{k,end}} - \underline{\Upsilon_{k,end}}. \quad (37)$$

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Thus, although the LMP is defined by (21) and (22) in all the three ISO-SOCM options, the primary difference lies in the manner in which the dual of the SOC constraint ($\Upsilon_{k,t}$) is calculated for the different time intervals within the optimization horizon. Note that the dual of the SOC constraint is a component of the LMP. When comparing options 1 and 3, the two options differ in the manner in which the shadow price is calculated for the end interval SOC constraint ($\Upsilon_{k,end}$). Note that the dual of any interval's SOC constraint ($\Upsilon_{k,t}$) within a multi-interval optimization horizon in options 1 and 3 is dependent on the dual of the end interval SOC constraint ($\Upsilon_{k,end}$) if the SOC limits are not binding in any of the intervals of the horizon. In addition, if the SOC limits are not binding in any of the intervals of the optimization horizon, option 1 sets $\Upsilon_{k,end}$ to equal $-\psi_k$, whereas option 3 sets $\Upsilon_{k,end}$ to equal $-c_k^{SOC}$. Options 1 and 3 can also be combined by penalizing deviations, in the objective, from the desired SOC level in the last interval. The dual for the end interval SOC constraint ($\Upsilon_{k,end}$) is the same for options 1 and 2 if a desired SOC level is provided in the last interval of the optimization horizon in option 2, i.e., if the last interval is included in the subset of time intervals with desired SOC levels ($end \in T^{SOC}$). Also, option 2 is unique in that the dual of any interval's SOC constraint ($\Upsilon_{k,t}$) within a multi-period optimization horizon is equal to (negative) summation of the dual of the desired SOC level constraint in that interval (if included, i.e., $\psi_{k,t:t\in T^{soc}}$) and all the *subsequent* intervals that have desired SOC levels provided if the SOC limits are not binding in any of the intervals of the horizon. In other words, $\Upsilon_{k,t} = -\sum_{t':t'\geq t,t'\in T^{soc}} \psi_{k,t'}$. Finally, it is worthwhile noting that in all the SOCM options, i.e., Self-SOCM, SOCM-Lite and ISO-SOCM, and otherwise (as a general practice) there are substantial benefits to solving the dual problem for the entire optimization horizon analogous to the primal problem; otherwise, MCPs may be set in an inefficient manner. This is an important feature to ensure ESRs (or/and other traditional resources) are able to set MCPs appropriately when marginal.

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